

Appunti Completi del Corso

Il Tuo Nome

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Contents

1		5
1.1	Introduction	5
1.2	Primer	5
1.2.1	Complex symmetric random variables, ∞ -modulations	5
1.3	Discrete entropy, proper signals	8
1.3.1	Entropy	8
1.3.2	Entropy rate	10
1.3.3	Differential entropy	10
1.4	Information divergence, mutual information	13
1.4.1	Power series expansions	14
1.4.2	Mutual information of a Gaussian scalar through an AWGN channel	15
2		17
2.1	MIMO Intuition	17
2.2	Mutual information of a gaussian vector through an AWGN channel . . .	18
2.2.1	Noise figure	18
2.3	Gaussian mutual info arbitrary const.	20
2.4	Mutual information of an encoded word	21
2.4.1	Theoretical treatment: results for generic $\mathbf{s}, \tilde{\mathbf{y}}$ (+for iid values and for a stationary channel)	21
2.5	Memoryless channel law with Gaussian noise	23
2.5.1	Capacity of a channel, information stability, spectral efficiency (+code- word error, information density)	23
2.6	BICM architecture (actually: Information loss when assuming bit inde- pendence, interleavers, soft decoding)	25
2.7	Spectral efficiency for finite-length codewords	30

Chapter 1

Week 13-19 October

1.1 Introduction

In most of the course we'll handle wireless communications, as they're much more difficult than wired communications. In this course we'll stay at the physical layer. We'll see:

- **Channel models:** taking radio communications and making a mathematical model for them. We'll find an accurate model and then we'll apply a tradeoff to it, simplifying it so that it is manageable.
- **OFDM:** Orthogonal Frequency Division Multiplexing.
- **MIMO:** we have multiple antennas at the transmitter and multiple antennas at the receiver; we'd like to see how much this improves communications. The book he selected this year handles fundamentals of MIMO communications. MIMO is the latest and greatest achievement, he chose a book which goes from the basics. We might have single user MIMO or even multiple user MIMO: we're communicating to multiple phones through multiple antennas.

We can ask ourselves: what is the best that we can do? This is what our approach will be, so that we can know the boundaries. We'll follow the slides from the author of the book, hopefully they'll be good. We'll also have exercises with solutions: we can look at them from cambridge.org, specifically from the page of the book.

1.2 Primer

The first chapter is a primer in information theory and MSE estimation; we'll go fast as it should summarize stuff we know, as mutual information between two variables. !!: the signals are always complex-valued and the constellations are always normalized to have unit power. ∞ -QAM: we increase the number of points making them denser inside a specific set.

1.2.1 Complex symmetric random variables, ∞ -modulations

We'll start from chapter 1 of the book (000): *primer of information theory and MSE*.

We should already know this stuff, but we'll cover it in order to have a common starting ground.

Prof's backstory moment: In the end, he studies **physical layer** and that's it.

We ask ourselves: “**what is information?** When can we say that someone is giving us information?”. *Professor begins ranting about information*: information is related to **probability!** The more unlikely something is, the more informative it is!

Today we'll talk about **mutual information** between two random variables: the information a variable can supply to another variable.

We'll talk about **channel capacity**: the capacity at which a channel can convey information; it answers to “how much information can I send through this channel, reliably?”. By reliably we mean that we can lower the probability of error as much as we want.

Binary coding and **decoding** are near optimum, but there can be other ways, which could be better: we'll show that binary is indeed **optimum**.

LMMSE is optimum when the signal and noise are **Gaussian**.

Let's start from **signals**; in the book, signals are normally **complex**, unless we specifically say otherwise. Normally **complex random variables** are **complexly symmetric**:

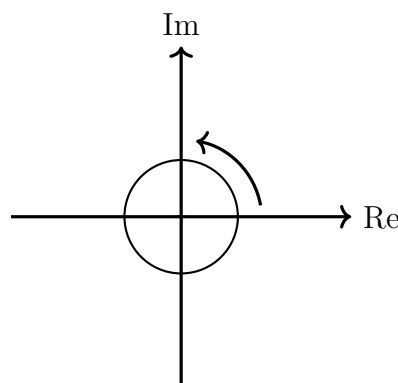
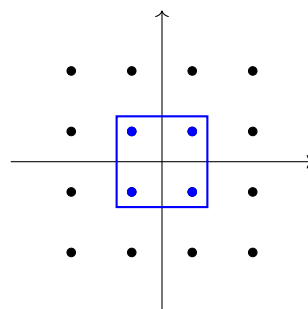


Figure 1.1: Rappresentazione del piano Im-Re con una PFD simmetrica centrale e freccia di rotazione.

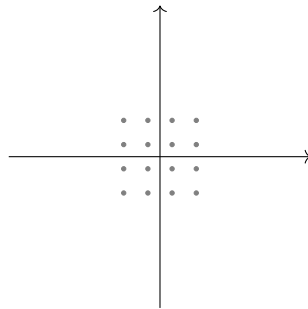
About **M-PSK** and **M-QAM**: M is usually a **power of 2**.

Let's see **16-QAM**:

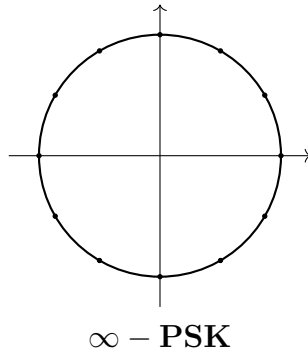


The power of the ones in blue is lower than the power of the other signals. In the book

they always keep the **same variance of the constellation**: the book squeezes the values so that the **power stays constant**.

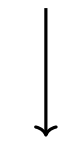


The book also has **inf-PSK** and **inf-QAM**: We take the **normal constellation** and we keep **adding points**.



It's a useful way to see what happens in the **asymptotic regime**. Also, the **PDF of a complex Gaussian variable exists** (see notes).

If σ^2 is the variance, the *Standard deviation* is $\sqrt{\sigma^2} = \sigma$



NOT $\pm\sigma^2$!

We'll try and be as precise as possible: there is no **negative standard deviation**! As an example:

$$f(x) = \sqrt{x} \in \mathbb{R} \text{ is def. only for } x \geq 0$$

$$x^2 = 16 \text{ has two sol.}$$

What about information? There have been many results, dating back to **Shannon**! (See pag 7 of the slides btw)

1.3 Discrete entropy, proper signals

1.3.1 Entropy

Entropy of a discrete-time random variable

$$\begin{aligned}\mathcal{H}(x) &= - \sum_x p_x(x) \log_2 p_x(x) \\ &= -\mathbb{E}[\log_2 p_x(x)].\end{aligned}$$

Joint entropy

$$\begin{aligned}\mathcal{H}(x_0, x_1) &= - \sum_{x_0} \sum_{x_1} p_{x_0 x_1}(x_0, x_1) \log_2 p_{x_0 x_1}(x_0, x_1) \\ &= -\mathbb{E}[\log_2 p_{x_0 x_1}(x_0, x_1)].\end{aligned}$$

Nonnegative function that quantifies the amount of uncertainty in discrete valued RV $\mathbf{x}$

Conditional entropy

$$\mathcal{H}(x|y) = - \sum_x \sum_y p_{xy}(x, y) \log_2 p_{x|y}(x|y).$$

Entropy connections

$$\mathcal{H}(x, y) = \mathcal{H}(x) + \mathcal{H}(y|x).$$

The entropy is used in statistical physics.

Joint entropy:

$$H(x_0, x_1) = \sum_{x_0} \sum_{x_1} p_{x_0, x_1}(x_0, x_1) \log_{1/2} p_{x_0, x_1}(x_0, x_1)$$

$\triangle$ **Diff. inform. can be negative**

Conditional entropy:

$$H(x, y) = \sum_x \sum_y p_{x,y}(x, y) \log_2(p_{x|y}(x|y))$$

From this, we can use the Bayes theorem to derive:

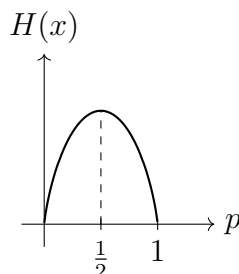
$$\begin{aligned}H(x, y) &= H(x|y) + H(y) \\ &= H(y|x) + H(x)\end{aligned}$$

Example: We have a Bernoulli variable

$$X = \{0, 1\} \quad P = \{p, 1 - p\}$$

$H(X)$ will be ≤ 1 , where the equality holds if $p = 1/2$.

If we plot it, we get:



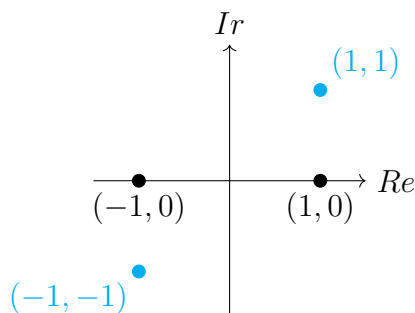
We get maximum information when we flip a ***fair*** coin!

If the coin always flips heads or tails, we get no information!

We could say that the messier the information, the most we get out of it.

Prof forgot stuff about signals: we're always talking about "proper" signals; the real and imaginary part of those numbers are uncorrelated!

Properness is preserved by affine transformations. An intuitive reason why we consider only proper complex numbers is that if we have real and imaginary parts of a random variable:



$(-1, -1)$ **This is a fake complex variable; we want to avoid these cases.**

To elaborate further:

Proper complex

A random variable (or a random process/random vector) is said to be proper complex if it satisfies **Properness**:

Properness

A complex random variable is said to be **proper** if

$$E[x^2] = E[x]^2$$

holds. Properness requires $\text{Re}[x], \text{Im}[x]$ to be uncorrelated and have the same variance.

A complex random vector is said to be **proper** if

$$E[\mathbf{x}\mathbf{x}^T] = E[\mathbf{x}]E[\mathbf{x}]^T$$

holds, which is equal to asking that $\text{Re}[\mathbf{x}], \text{Im}[\mathbf{x}]$ have identical Covariance matrices and cross-covariance equal to 0.

Properness is preserved in affine transformations!

1.3.2 Entropy rate

Let's say we have a sequence of discrete random variables indexed by time, $x_0, \dots, x_{N-1}$. If they are **IID**, then the entropy of the whole process grows linearly with N at a rate $\mathcal{H}(x_0)$.

In general, a process grows according to the so-called *entropy rate*

$$\mathcal{H} = \lim_{N \rightarrow +\infty} \frac{1}{N} \mathcal{H}(x_0, \dots, x_{N-1}). \quad (1.32)$$

If the process is **stationary**, we can show that the entropy rate is equal to

$$\mathcal{H} = \lim_{N \rightarrow +\infty} \mathcal{H}(x_N | x_0, \dots, x_{N-1}). \quad (1.33)$$

All of this can be extended for continuous processes to differential entropy.

A process is said to be **nonregular** if its present value is perfectly predictable from noiseless observations of the entire past, while the process is **regular** (which happens if $\mathfrak{h} > -\infty$) if its present value cannot be perfectly predicted from noiseless observations of the entire past.

1.3.3 Differential entropy

Differential entropy:

$$h(x) = \int f_x(x) \log_{1/2} f_x(x) dx = \mathbb{E} [\log_{1/2} f_x(x)]$$

For vectors:

$$h(\underline{x}) = \mathbb{E} [\log_{1/2} f_{\underline{x}}(\underline{x})]$$

The differential entropy rate does not measure the information contained in a random variable. If we only have a random variable, we only have a shot. In order to consider a **rate**, we consider a **random process**: in this case, we have a random variable every certain amount of time: we can say that we have a certain amount of information per second!

A binary random process with $p = 0.5$ which gives a value every second, gives 1 bit/s.

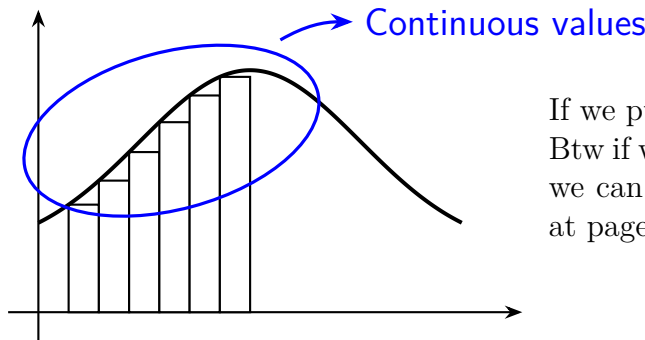
So the rate does not measure the information contained in a random variable. We also need to keep in mind that the differential entropy can be negative.

It's easy to prove that a **Gaussian distribution maximizes the differential entropy**, for a fixed variance.

The differential entropy of a b bit quantization of a random variable x is $h(x) + b$ (what happens if we quantize more and more finely?)

If we discretize a variable in b bits, then what we obtain is another random variable. What if we want to compute the differential entropy of that random variable?

Why do we use differential entropy even though we discretized? We just elongate it:



If we push b to ∞ , it diverges!

Btw if we want a reference for information theory, we can use the book "Cover & Thomas" (this is at page 247).

Pag 10 of the slides: how to calculate the differential entropy of a Gaussian complex vector.

Example 1.6 (Differential entropy of a complex Gaussian vector)

Let $\mathbf{x} \sim \mathcal{N}_c(\boldsymbol{\mu}, \mathbf{R})$. From (C.15) and (1.22).

$$h(\mathbf{x}) = -\mathbb{E}[\log_2 f_{\mathbf{x}}(\mathbf{x})] \quad (1.23)$$

$$= \log_2 \det(\pi \mathbf{R}) + \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})^* \mathbf{R}^{-1} (\mathbf{x} - \boldsymbol{\mu})] \log_2 e \quad (1.24)$$

$$= \log_2 \det(\pi \mathbf{R}) + \text{tr}(\mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})^* \mathbf{R}^{-1} (\mathbf{x} - \boldsymbol{\mu})]) \log_2 e \quad (1.25)$$

$$= \log_2 \det(\pi \mathbf{R}) + \text{tr}(\mathbb{E}[\mathbf{R}^{-1} (\mathbf{x} - \boldsymbol{\mu}) (\mathbf{x} - \boldsymbol{\mu})^*]) \log_2 e \quad (1.26)$$

$$= \log_2 \det(\pi \mathbf{R}) + \text{tr}(\underbrace{\mathbf{R}^{-1} \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu}) (\mathbf{x} - \boldsymbol{\mu})^*]}_{\mathbf{R}}) \log_2 e \quad (1.27)$$

$$= \log_2 \det(\pi \mathbf{R}) + \text{tr}(\mathbf{I}) \log_2 e \quad (1.28)$$

$$= \log_2 \det(\pi e \mathbf{R}), \quad (1.29)$$

where in (1.25) we used the fact that a scalar equals its trace, while in (1.26) we invoked the commutative property in (B.26).

Note: $\det(\exp(\mathbf{A})) = \exp(\text{tr}(\mathbf{A}))$

Example 1.5 (Differential entropy of a complex Gaussian scalar)

Let $x \sim \mathcal{N}_C(\mu, \sigma^2)$. Invoking the PDF in (C.14),

$$h(x) = \mathbb{E} \left[\frac{|x - \mu|^2}{\sigma^2} \log_2 e + \log_2(\pi \sigma^2) \right] = \log_2(\pi e \sigma^2).$$

(Scalar result)

Equivalent scalar result

Professor freaks out over the fact that we don't know what the trace function is. The trace of a matrix is the sum of the diagonals. If we want to compute the sum of the eigenvalues, we can just compute the trace of a matrix.

btw $\det(\exp(\underline{A})) = \exp(\text{tr}(\underline{A}))$

So we have that

$$\begin{aligned} h(\underline{x}) &= \log_2(\det(\pi e \underline{R})) \\ h(x) &= \log_2(\pi e \sigma^2) \end{aligned}$$

At least not for a Gaussian R.V.!

The differential entropy DOES NOT depend on the mean of the variable!

Another very important thing is that the differential entropy is a property of the distribution of a random variable, not of the random variable itself. Two random variables having the same distribution will have the same random variable.

$$\begin{array}{ccc} R.V. \text{ is a map} & \Omega & \longrightarrow \mathcal{S} \\ & \downarrow & \downarrow \\ & \text{events} & \text{probability} \end{array}$$

Elaborating a bit on the computation of the differential entropy: we have that

$$h(\mathbf{x}) = -\mathbb{E}[\log_2 f_{\mathbf{X}}(\mathbf{x})] \tag{1.23}$$

$$\begin{aligned} &= -\mathbb{E} \left[\log_2 \left(\frac{1}{\det(\pi \mathbf{R}_x)} \exp(-(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)) \right) \right] \\ &= \log_2(\det(\pi \mathbf{R}_x)) - \mathbb{E} \left[\frac{\ln(\exp(-(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)))}{\ln 2} \right] \\ &= \log_2(\det(\pi \mathbf{R}_x)) - \mathbb{E} \left[\frac{-(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)}{\ln 2} \right] \\ &= \log_2(\det(\pi \mathbf{R}_x)) + \mathbb{E} [(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)] \log_2 e \end{aligned} \tag{1.24}$$

and now we observe that $(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)$ actually is a scalar (given that $\mathbf{x}$ is a $N \times 1$ vector, we're multiplying $(1 \times N) \cdot (N \times N) \cdot (N \times 1) \rightarrow 1$): we can then exploit the fact that a scalar is equal to its trace together with the fact that the trace is a scalar operation to get

$$= \log_2(\det(\pi \mathbf{R}_x)) + \text{Tr} [\mathbb{E} [(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)]] \log_2 e \quad (1.25)$$

$$= \log_2(\det(\pi \mathbf{R}_x)) + \mathbb{E} [\text{Tr} [(\mathbf{X} - \boldsymbol{\mu}_x)^* \mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x)]] \log_2 e \quad \text{Linearity}$$

$$= \log_2(\det(\pi \mathbf{R}_x)) + \mathbb{E} [\text{Tr} [\mathbf{R}_x^{-1} (\mathbf{X} - \boldsymbol{\mu}_x) (\mathbf{X} - \boldsymbol{\mu}_x)^*]] \log_2 e \quad \text{Ciclicity}$$

$$= \log_2(\det(\pi \mathbf{R}_x)) + \text{Tr} [\mathbf{R}_x^{-1} \mathbf{R}_x] \log_2 e$$

$$= \log_2(\det(\pi \mathbf{R}_x)) + \text{Tr} [\mathbf{I}] \log_2 e \quad (1.28)$$

if then we consider $\mathbf{R}_x, \mathbf{I}$ as $N \times N$ matrices (which they are, this is just needed to say that N is their measure), we'll have

$$= \log_2(\det(\pi \mathbf{R}_x)) + N \log_2 e$$

$$= \log_2(e^N \det(\pi \mathbf{R}_x))$$

and, finally, for a matrix $\mathbf{A}$ of dimensionality $N \times N$ it holds that

$$a^N \det(\mathbf{A}) = \det(a\mathbf{A}),$$

so in our situation this is

$$= \log_2(\det(\pi e \mathbf{R}_x)). \quad (1.29)$$

Whew, that was a lot!

Finally, we can simplify this for the scalar case as the determinant of a scalar is the scalar itself, getting that

$$h(x) = \log_2(\pi e \sigma^2).$$

| Btw the B.26 relation that they mentioned is

$$\text{tr}(\mathbf{A}\mathbf{B}) = \text{tr}(\mathbf{B}\mathbf{A}), \quad (\text{B.26})$$

which was here applied to $\mathbf{A} = (\mathbf{x} - \boldsymbol{\mu})^* \mathbf{R}^{-1}$, $\mathbf{B} = (\mathbf{x} - \boldsymbol{\mu})$.

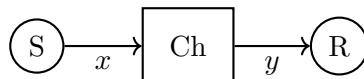
1.4 Information divergence, mutual information

We have that the Kullback-Leibler divergence is defined as

$$\mathcal{D}(p||q) = \begin{cases} \sum_x p(x) \log_2 \left(\frac{p(x)}{q(x)} \right) & \text{discrete } x \\ \int p(x) \log_2 \left(\frac{p(x)}{q(x)} \right) & \text{continuous } x \end{cases} \quad (1.34+1.36)$$

this is also called the *relative entropy*.

Another way to measure such thing is the Earth-Mover distance.



The mutual information quantifies the reduction in uncertainty between two random variables if one of them is observed.

Discrete:

$$\begin{aligned} I(s; y) &= H(s) - H(s|y) = H(y) - H(y|s) \\ &\quad \swarrow \text{Between } s \text{ and } y \\ &= D(p_{sy} || p_s p_y) \\ &= \sum_s \sum_y p_{sy}(s, y) \log_2 \left(\frac{p_{sy}(s, y)}{p_s(s) p_y(y)} \right) \end{aligned}$$

Cont:

$$\begin{aligned} &= \mathbb{E}[\text{information density}] \equiv \mathbb{E}[i(s; y)] \\ I(s; y) &= h(s) - h(s|y) = h(y) - h(y|s) \\ &= D(f_{sy} || f_s f_y) \\ &= \iint f_{sy}(s, y) \log_2 \left(\frac{f_{sy}(s, y)}{f_s(s) f_y(y)} \right) ds dy \end{aligned}$$

He talked about the **Landau symbols**; he didn't say anything relevant really.

1.4.1 Power series expansions

Useful means to approximate functions especially low or high SNR.

- **Zero order expansion about 0**

$$f(x) = f(0) + \mathcal{O}(x)$$

- **First order expansion about 0**

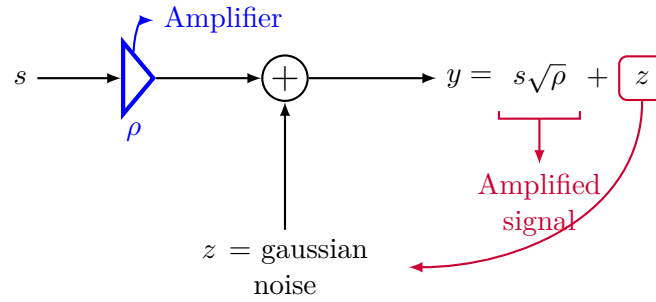
$$f(x) = f(0) + f'(0)x + \mathcal{O}(x^2)$$

- **First order expansion about Infinity**

$$f(1/x) = f(0) + \frac{f'(0)}{x} + \mathcal{O}\left(\frac{1}{x^2}\right)$$

Mathematica snippet:

```
In[11]:= Series[f[x], {x, 0, 2}]
Out[11]= f[0] + f'[0]x + 1/2 f''[0]x^2 + O[x]^3
```

Important! → SL. 16

We can ask ourselves: how much is the gaussian noise corrupting what I'm sending?

We amplify ρ (which is $\sqrt{\rho}$ in power) $\Rightarrow$ we get an ampl. of $\sqrt{\rho}$ on power!

1.4.2 Mutual information of a Gaussian scalar through an AWGN channel

We were calculating the **mutual information** for a Gaussian complex scalar.

We talked about the relation $y = \sqrt{\rho}s + z$. We have that s, z are **independent**. It might seem weird that we're putting noise in a channel, but it will make sense later.

Our constellation input always will have $\sigma^2 = 1$, such that $s \sim \mathcal{N}_{\mathcal{C}}(0, 1)$; we'll also have a noise $z \sim \mathcal{N}_{\mathcal{C}}(0, 1)$. We have that y is made of a sum of Gaussian independent random variables, and will thus be a Gaussian random variable, with a variance given by a sum of variances: we'll have

- $y \sim \mathcal{N}_{\mathcal{C}}(0, 1 + \rho)$
- $y|s \sim \mathcal{N}_{\mathcal{C}}(\sqrt{\rho}s, 1)$: this is y with s being fixed; we fix the $\sqrt{\rho}s$ in y 's equation. We now want to calculate the mutual information between the input and the output of the signal.

We know the differential entropy of a Gaussian scalar and we know that

$$\begin{aligned}
 \mathcal{I}(\rho) &= I(s; \sqrt{\rho}s + z) \\
 &= h(\sqrt{\rho}s + z) - h(\sqrt{\rho}s + z|s) \\
 &= h(\sqrt{\rho}s + z) - h(z) \\
 &= \log_2(\pi e(1 + \rho)) - \log_2(\pi e) \\
 &= \log_2(1 + \rho).
 \end{aligned} \tag{1.51}$$

Relevant comments:

- In $h(\sqrt{\rho}s + z|s) = h(z)$, we know that we've already **fixed** s ; s multiplied by whichever scalar won't bring any extra information than z , as we already know what s is; for this reason, the differential entropies are the same.

If we then consider $\rho \rightarrow 0$, we have

$$\log_e(1+x) \xrightarrow{x \rightarrow 0} x - \frac{1}{2}x^2 + \mathcal{O}(x^3)$$

So we need to **convert to base e** and we'll then be able to use the expansion:

$$\log_2(1+\rho) = \left(\rho - \frac{1}{2}\rho^2\right) \log_2 e + \mathcal{O}(\rho^3)$$

And **for large ρ** we'll then have

$$\log_2(1+\rho) \xrightarrow{\rho \rightarrow \infty} \log_2(\rho) + \mathcal{O}(1/\rho)$$

■ **Note**

This is **crucially important** as it allows to model an AWGN channel! It allows us to define the **capacity** as

$$\frac{1}{2} \log(1 + \text{SNR})$$

(we don't have $1/2$ since this is complex; we'd have $1/2$ if it wasn't complex!).

Chapter 2

Chapter 2 Week 27 October - 2 November

2.1 MIMO Intuition

We'll use random vectors in order to model MIMO communications Our book is **MIMO**! For a long time we had a setup with a transmitter and a receiver.

People asked: what can we do in order to increase the **spectral efficiency** of our signals? The spectral efficiency is how many **bit/s/Hz** we can squeeze.

The spectral efficiency we've reached is between **1 and 5 bit/s/Hz**; we generally have **4 bit/s/Hz**.

$B \cdot \nu$ is how many bits we can send (B = bandwidth, ν = spectral efficiency).

What if we have **two antennas** at the transmitter and **two at the receiver**? It's been shown that by doing such, **it's like having twice the bandwidth**: this is the whole concept of **MIMO**.

Of course we have to pay **double**.

With two antennas we're transmitting two symbols, aka a **vector**:

$$\begin{bmatrix} T_x \\ S_1 \\ S_2 \end{bmatrix}$$

And the received signal vector is:

$$\begin{bmatrix} L_{RC} \\ y_1 \\ y_2 \end{bmatrix} = A \begin{bmatrix} S_1 \\ S_2 \end{bmatrix} + \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

We thus need to use **vectors**, we can't stay with **scalars**!

2.2 Mutual information of a gaussian vector through an AWGN channel (+figure of noise)

Let's say we have a vector $\mathbf{s}$ and a matrix $\mathbf{A}$ (which is needed since every signal is communicating with each antenna!): **we're now in the MIMO setting**.

We have a **Gaussian vector** $\mathbf{s}$, a **matrix** $\mathbf{A}$ and a **noise Gaussian vector** $\mathbf{z}$; we'll have an output

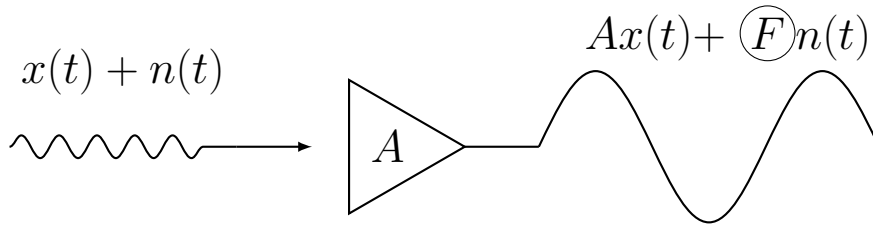
$$\mathbf{y} = \sqrt{P_A} \mathbf{A} \mathbf{s} + \mathbf{z},$$

where $\mathbf{A}$ is considered **deterministic** (we'll see it's not really true), $\mathbf{z}$ is the **noise**, which originates from **thermal noise: electrons are not still**; the more we increase the temperature, the more **electrons move**; we want to transmit a certain voltage, but we'll measure a certain fluctuating value added to our transmitted signal: the **fluctuating addition comes from thermal noise!**

2.2.1 Noise figure

▽ Noise figure

What about **noise figure**? When we amplify a signal, do we pay a price? We're amplifying the noise as well, but we might amplify the noise more: the amplifier will add its own noise to the noise, which is F , the noise figure:



We now want to calculate the mutual information between source and output; the idea is the same, only with matrices and vectors; we have

$$\mathcal{I}(\rho) = I(\mathbf{s}; \sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) \quad (1.74)$$

$$= h(\sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) - h(\sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z} | \mathbf{s}) \quad (1.75)$$

$$= h(\sqrt{\rho} \mathbf{A} \mathbf{s} + \mathbf{z}) - h(\mathbf{z}) \quad (1.76)$$

$$= \log_2 \det(\pi e(\rho \mathbf{A} \mathbf{R}_s \mathbf{A}^* + \mathbf{R}_z)) - \log_2 \det(\pi e \mathbf{R}_z) \quad (1.77)$$

$$= \log_2 \det(\mathbf{I} + \rho \mathbf{A} \mathbf{R}_s \mathbf{A}^* \mathbf{R}_z^{-1}), \quad (1.78)$$

with (1.77) following from the definition of **differential entropy of a gaussian vector**. The correlation matrix of $\sqrt{\rho} \mathbf{A} \mathbf{s}$ is $\rho \mathbf{A} \mathbf{R}_s \mathbf{A}^*$; we then have a sum, so we'll have $\mathbf{R}_z$. The fact that in the last passage (1.78) we have $\mathbf{R}_z$ at the denominator, means we'll be multiplying by $\mathbf{R}_z^{-1}$; this follows from the fact that

$$\frac{\det(\mathbf{A})}{\det(\mathbf{B})} = \det(\mathbf{A}\mathbf{B}^{-1}),$$

here applied with $\mathbf{A} = \rho\mathbf{A}\mathbf{R}_s\mathbf{A}^* + \mathbf{R}_z$ and $\mathbf{B} = \mathbf{R}_z$.

⊠ How do we know that $\sqrt{\rho}\mathbf{A}\mathbf{s} + \mathbf{z}$ is Gaussian?

I think the idea is that we can consider $\sqrt{\rho}\mathbf{A}\mathbf{s} + \mathbf{z}$ as a multiplication between the matrix

$$[\sqrt{\rho}\mathbf{A} \quad 1]$$

and the vector

$$\begin{bmatrix} \mathbf{s} \\ \mathbf{z} \end{bmatrix},$$

which will result in a random vector with distribution

$$\sim \mathcal{N}_{\mathbb{C}} \left([\sqrt{\rho}\mathbf{A} \quad 1] \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix}, [\sqrt{\rho}\mathbf{A} \quad 1] \cdot \begin{bmatrix} \mathbf{R}_s & 0 \\ 0 & \mathbf{R}_z \end{bmatrix} \cdot \begin{bmatrix} \sqrt{\rho}\mathbf{A}^* \\ 1 \end{bmatrix} \right),$$

according to

Transformation of a Gaussian vector

If we have:

- A Gaussian vector $\mathbf{x} \sim \mathcal{N}_{\mathbb{C}}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ of dimensions $N \times 1$,
- a matrix $\mathbf{A}$ of dimensions $M \times N$
- a vector $\mathbf{b}$ of dimensions $M \times 1$

we have that the vector

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$$

will be distributed according to

$$\mathbf{y} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^*).$$

► Extension of **Uni generale**.

Derivative of logarithm of determinant

In order to go on, we consider the following relations:

$$\begin{aligned} \left. \frac{\partial}{\partial \rho} \log_e \det(\mathbf{I} + \rho\mathbf{B}) \right|_{\rho=0} &= \text{tr}(\mathbf{B}) \\ \left. \frac{\partial^2}{\partial \rho^2} \log_e \det(\mathbf{I} + \rho\mathbf{B}) \right|_{\rho=0} &= -\text{tr}(\mathbf{B}^2) \end{aligned}$$

We can then apply some linear algebra: we'll get that for $\rho \rightarrow 0$ follows

$$\mathcal{I}(\rho) = \left[\text{tr}(\mathbf{A}\mathbf{R}_s\mathbf{A}^*\mathbf{R}_z^{-1})\rho - \frac{1}{2}\text{tr}((\mathbf{A}\mathbf{R}_s\mathbf{A}^*\mathbf{R}_z^{-1})^2)\rho^2 \right] \log_2 e + o(\rho^2)$$

About this passage: it all depends on the fact that $\log \det$ is a **convex function**! We know the derivative of $\log \det$ and we're applying it to the fact that we're summing the eigenvalues: that's how we get the **trace**!

Page **74** of a linear algebra book he recommended

This is a very useful result!

If N_s, N_y are the dimensions of the vectors, for $\rho \rightarrow +\infty$ we get the result

$$\mathcal{I}(\rho) = \min(N_s, N_y) \log_2 \rho + \log_2 \det(\mathbf{A}\mathbf{R}_s\mathbf{A}^*\mathbf{R}_z^{-1}) + O\left(\frac{1}{\rho}\right)$$

Professor said that this equation has a minor detail which is **wrong**, proceeded into not telling us what it was :D

2.3 Gaussian mutual information for an arbitrary constellation

Let's see the Gaussian mutual information for an arbitrary constellation.

Let $\mathbf{s}$ be a zero-mean unit-variance discrete random variable taking values in $\mathbf{s}_0, \dots, \mathbf{s}_{M-1}$ with probabilities $p_0, \dots, p_{M-1}$. This subsumes **M-PSK**, **M-QAM**, and any other discrete constellation. The PDF of $\mathbf{y} = \sqrt{\rho}\mathbf{s} + \mathbf{z}$ equals

$$f_{\mathbf{y}}(y) = \frac{1}{\sqrt{2\pi}} \sum_{m=0}^{M-1} p_m e^{-|y - \sqrt{\rho}\mathbf{s}_m|^2} \quad (1.65v)$$

whereas $y|\mathbf{s} \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}\mathbf{s}, 1)$. Thus,

$$\mathcal{I}^{\text{aw}}(\rho) = I(\mathbf{s}; \sqrt{\rho}\mathbf{s} + \mathbf{z}) \quad (1.66)$$

$$= - \int f_{\mathbf{y}}(y) \log_2 f_{\mathbf{y}}(y) dy - \log_2(\pi e) \quad (1.67)$$

with integration over the complex plane.

$$\mathcal{I}_{\text{aw}}^{\text{aw}}(\rho) = \log_2 M - \epsilon \quad \text{Large a first order expansion} \quad \log_2 e = \frac{d_{\min}^2}{4} \rho + o(\rho)$$

dove

$$d_{\min} = \min_{k \neq l} |\mathbf{s}_k - \mathbf{s}_l|$$

Comments:

- In order to obtain (1.65) we used

$$f_{\mathbf{y}}(y) = \sum_{m=0}^{M-1} f_{\mathbf{y}|\mathbf{s}}(y|\mathbf{s}_m) p_{\mathbf{s}}(\mathbf{s}_m),$$

with $p_{\mathbf{s}}(\mathbf{s}_m) = p_m$.

- (1.67) is just $h(\mathbf{y}) - h(\mathbf{y}|\mathbf{s})$, considering that $\mathbf{y}|\mathbf{s}$ is a **Gaussian RV with variance 1** and thus $h(\mathbf{y}|\mathbf{s}) = h(\mathbf{z}) = \log_2(\pi e)$.
- We don't see σ^2 since it's $\sigma_z^2 = 1$!
- ϵ is first defined here

(1.66) is the definition of mutual information. Solving the integral is pretty difficult.

What professor wants us to remember though is that

$$\mathcal{I}^{\text{aw-ary}}(\rho) \xrightarrow{\rho \rightarrow +\infty} \log_2(M) - \epsilon,$$

with

$$\log_2(\epsilon) = \frac{d_{\min}^2}{4} \rho + o(\rho)$$

intuitively, a constellation with M points asymptotically gives $\log_2(M)$ bits of information per channel use; of course this is not the case and we need to lower that term using ϵ , which brings us from the **best case scenario** to a more realistic case.

2.4 Mutual information of an encoded word

2.4.1 Theoretical treatment: results for generic $\mathbf{s}, \tilde{\mathbf{y}}$ (+for iid values and for a stationary channel)

We have an abstraction of the communication link:

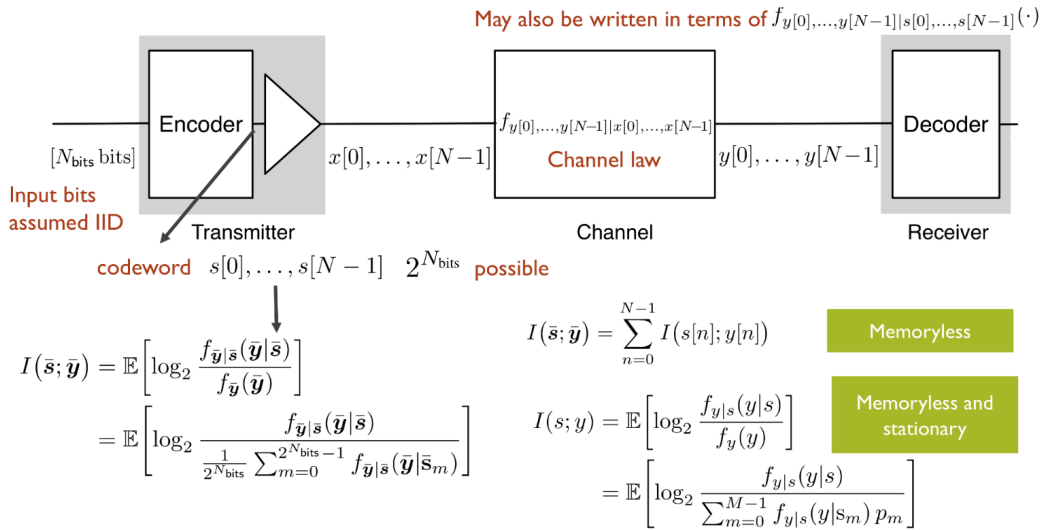


Figure 2.1: Abstraction of the communication link

We have an encoder, which maps a signal to a codebook with $2^{N_{\text{bits}}}$ words. We send a codeword of N bits $x[n], n \in [0, N-1]$. We then send our signal through the channel, which is characterized by **noise**, **amplification**, $\dots$; we'll receive a whole word $y[n], n \in [0, N-1]$. If we want to calculate the **mutual information between input words and output words**, we need to use the **conditioned information function**, not the joint one: we know the probability of receiving a certain message $\mathbf{y} = [y[0], \dots, y[N-1]]$ (as it is $f_{\mathbf{y}}$) when sending a certain $\mathbf{s} = [s[0], \dots, s[N-1]]$ and we need to work with that!

If we assume that the various symbols are **equiprobable**, we find that

$$f_{\mathbf{y}}(\mathbf{Y}) = \sum_{m=0}^{2^{N_{\text{bits}}}-1} f_{\mathbf{y}|\mathbf{s}}(\mathbf{Y}|\mathbf{s}_m) p_{\mathbf{s}}(\mathbf{s}_m) \stackrel{\text{equiprobable } \mathbf{s}_m}{=} \sum_{m=0}^{2^{N_{\text{bits}}}-1} f_{\mathbf{y}|\mathbf{s}}(\mathbf{Y}|\mathbf{s}_m) \frac{1}{2^{N_{\text{bits}}}}$$

and this enables us to write

$$I(\mathbf{s}; \mathbf{y}) = \mathcal{H}(\mathbf{y}) - \mathcal{H}(\mathbf{y}|\mathbf{s}) \quad (1.88)$$

$$\begin{aligned} &= \mathbb{E} \left[\log_2 \left(\frac{f_{\mathbf{y}|\mathbf{s}}(\mathbf{y}|\mathbf{s})}{f_{\mathbf{y}}(\mathbf{y})} \right) \right] \\ &= \mathbb{E} \left[\log_2 \left(\frac{f_{\mathbf{y}|\mathbf{s}}(\mathbf{y}|\mathbf{s})}{\frac{1}{2^{N_{\text{bits}}}} \sum_{m=0}^{2^{N_{\text{bits}}}-1} f_{\mathbf{y}|\mathbf{s}}(\mathbf{Y}|\mathbf{s}_m)} \right) \right]. \end{aligned} \quad (1.89)$$

If the channel is **memory-less** (which might not be true: $y[1]$ might depend on $y[0]$ etc etc), the mutual information of input and output is the **sum of mutual information of every couple of input and output**: we don't have vectors anymore (and we like this!).

If this is the case, the channel law will factor out as

$$f_{\mathbf{y}|\mathbf{s}}(\mathbf{y}|\mathbf{s}) = \prod_{n=0}^{N-1} f_{y[n]|\mathbf{s}[n]}(y[n]|\mathbf{s}[n]).$$

with

$$I(s[n]; y[n]) = \mathbb{E} \left[\log_2 \frac{f_{y[n]|\mathbf{s}[n]}(y[n]|\mathbf{s}[n])}{f_{y[n]}(y[n])} \right]. \quad (1.93)$$

Hidden message

Looks like one hidden message is that if the conditioned probability factorizes, we can easily rewrite the mutual information as a summation. This follows from the fact that if the single elements are iid, then the expected value factorizes as well.

If the channel is also **stationary**, we can simplify further, as all the symbols will have the same statistics: $s \equiv s[n] \forall n$ and $y \equiv y[n] \forall n$ will hold, and can write (1.93) as

$$I(s; y) = \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{f_y(y)} \right] \quad (1.94)$$

$$= \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{\sum_{m=0}^{M-1} f_{y|s}(y|s_m) p_m} \right], \quad (1.95)$$

with p_m being the probability that a symbol s is s_m ; we can of course further simplify this writing if we assume equiprobability, $p_m = 1/M$.

This comes from the fact that any output does not store any information about any input other than the one which corresponds to it:

$$I(s[k]; y[n]) = \begin{cases} 0 & k \neq n \\ \text{Something} & \text{otherwise} \end{cases}.$$

2.5 Memoryless channel law with Gaussian noise

Let's see an example; if our output is defined as

$$y[n] = \sqrt{\rho} s[n] + z[n] \quad n = 0, \dots, N-1 \quad (1.96)$$

and we have that $z[0], \dots, z[N-1]$ are IID with $z \sim \mathcal{N}_{\mathbb{C}}(0, 1)$ the channel law will be the same as z , now centered in mean in the same value as the transmitted s :

$$f_{y|s}(y|s) = \frac{1}{\pi} e^{-|y - \sqrt{\rho} s|^2}. \quad (1.97)$$

The channel law is the probability that we receive y when we're transmitting s .

2.5.1 Capacity of a channel, information stability, spectral efficiency (+codeword error, information density)

We come to an extra important concept: **capacity**. We can express it in different ways; we'll see **bit/s**; when we'll put the symbols one after the other.

Ideally we must study the book, in order to fully grasp these concepts

The **codeword error** probability for equiprobable words of length N is

$$p_e = \frac{1}{2^N} \sum_{m=0}^{2^N-1} P[\hat{w} \neq m | w = m],$$

where w is the transmitted word and $\hat{w}$ is the received word.

Capacity (bits/symbol)

The capacity C in bits per symbol is the highest rate that can be achieved such that for large values of N we have

$$p_e \xrightarrow{N \rightarrow +\infty} 0.$$

The **information density** between an input word $\bar{s}$ and an output word $\bar{y}$ is defined as

$$i(\bar{s}; \bar{y}) = \log_2 \left(\frac{f_{\bar{s}, \bar{y}}(\bar{S}, \bar{Y})}{f_{\bar{s}}(\bar{S})f_{\bar{y}}(\bar{Y})} \right) .$$

If the **channel** is **information stable**, then we have that

$$\lim_{N \rightarrow +\infty} \frac{1}{N} i(\bar{s}, \bar{y}) = \lim_{N \rightarrow +\infty} \frac{1}{N} \mathbb{E}[i(\bar{s}, \bar{y})] \stackrel{*}{=} \lim_{N \rightarrow +\infty} \frac{1}{N} I(\bar{s}, \bar{y}) .$$

Professor's definition of information stable

A channel is information stable if the mutual information does not vary if we jump in the future or past; it's not taking the first symbols and conveying less information than the symbols coming in later.

This relation is saying that the information conveyed by $y[0], \dots, y[N-1]$ about $s[0], \dots, s[N-1]$ is invariant, provided that N is large enough

*: note that, by **definition**,

$$\mathbb{E}[i(s; y)] = I(s; y) .$$

Sufficient condition for information stability

A sufficient condition for a channel to be information stable is that **the channel is stationary and ergodic** (ref)

Ergodicity in mean

We have a random process $x(n)$ and we know that $\mathbb{E}[x(n)] = B$.

A signal is **ergodic in mean** if

$$\lim_{N \rightarrow +\infty} \sum_{n=0}^{N-1} \frac{x(n)}{N} = B :$$

the average in time is the same as the average taken in the probability space

If the channel is stationary and ergodic, then

$$C = \max_{\text{signal constraints}} \left(\lim_{N \rightarrow +\infty} \frac{1}{N} I(\bar{s}, \bar{y}) \right) .$$

The maximization is over the joint distribution of the unit-power codeword symbols $s[0], \dots, s[N-1]$.

Zero mean signals

The mean has no influence over the mutual information (and thus the capacity), but it does influence the average power spent for the transmission; for this reason, from now onwards, we'll only consider zero-mean signals.

We want the codeword error probability to go to zero for long words. We define an **information stable channel**; we then have the information density: if it's information stable, the information density must become the mutual information for large N s. The capacity is the **maximum mutual information which we can achieve by changing the distribution of the input signal**. We have very long codewords and we want the probability of error on the codeword to be as low as we want.

If we have a pipe, the capacity is how much water we can push through without losing any water

In order to reach the capacity, what we can do is changing the input distribution of the symbols.

If the channel is **static, memoryless and ergodic**, then we simply have

$$C = \max_{\text{signal constraints}} I(s; y).$$

The bitrate R at which we can reliably communicate with a period T satisfies $R \leq C/T$. From the sampling theorem we have $1/T \leq B$, where B is the passband bandwidth of our transmission. By joining the two conditions, we obtain that

$$\frac{R}{B} \leq C \leftrightarrow \nu \leq C,$$

ν , spectral efficiency

where C uses the alternative measure bit/s/Hz and where the equality is reached asymptotically.

This is telling us that the capacity is not only the maximum bitrate, but also the maximum spectral capacity which we can reach.

The capacity involves searching in all the possible distributions, we'll always stay underneath it, and that's something we always need to take into account!

2.6 BICM architecture (actually: Information loss when assuming bit independence, interleavers, soft decoding)

From the book: In paragraphs **before 1.5.4**, the book showed that using **binary encoding** and **constellation mapping** together yields no loss of optimality. We can ask ourselves whether this also holds when we combine **soft demapping** and **binary decoding**. In order to investigate this, we consider a stationary and memoryless channel as well as an M -point equiprobable constellation.

In this setting, codewords with IID entries are optimum and thus bits mapped to distinct symbols can be taken to be independent. However, **the channel does introduce dependencies among same-symbol bits**. Even with the coded bits produced by the binary encoder being statistically independent (as we know from *past paragraphs*), after demapping at the receiver dependencies do exist among the soft values for bits that traveled on the same symbol. Unaware, a binary decoder designed for IID bits ignores these dependencies and regards the channel as being memoryless, not only at a symbol level but further at a bit level [91]. Let us see **how much information can be recovered under this premise**.

First of all, we look at the decoder counterpart of (1.94), (1.95), if the channel is also stationary, we can simplify further, as all the symbols will have the same statistics: $s \equiv s[n] \forall n$ and $y \equiv y[n] \forall n$ will hold, and can write (1.93) as

$$I(s; y) = \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{f_y(y)} \right] \quad (1.94)$$

$$= \mathbb{E} \left[\log_2 \frac{f_{y|s}(y|s)}{\sum_{m=0}^{M-1} f_{y|s}(y|s_m) p_m} \right] \quad (1.95)$$

with p_m being the probability that a symbol s is s_m ; we can of course further simplify this writing if we assume equiprobability, $p_m = 1/M$, which is

$$I_{\text{aw}}^{\text{BICM}} = \sum_{k=0}^{m_k-1} I(\mathbf{b}_k; \mathbf{y})$$

after much hard work, we find that it is

$$I_{\text{aw}}^{\text{BICM}} = \mathbb{E}_{\mathbf{y}} \left[\sum_{k=1}^{m_k} \mathbb{E}_{\mathbf{s}} \left[\log_2 \frac{\sum_{i=0}^{2^{m_k}-1} \mathbb{E}_{\mathbf{s}_{\setminus k}} [f_{\mathbf{y}|\mathbf{s}}(\mathbf{y}|\mathbf{s}_i)]}{\sum_{i=0}^{2^{m_k}-1} f_{\mathbf{y}|\mathbf{s}}(\mathbf{y}|\mathbf{s}_i)} \right] \right] \quad (1.137)$$

where $\mathbf{s}[k]$ are the subsets of constellation points where k -th bit is either 0 or 1; just to give an example:

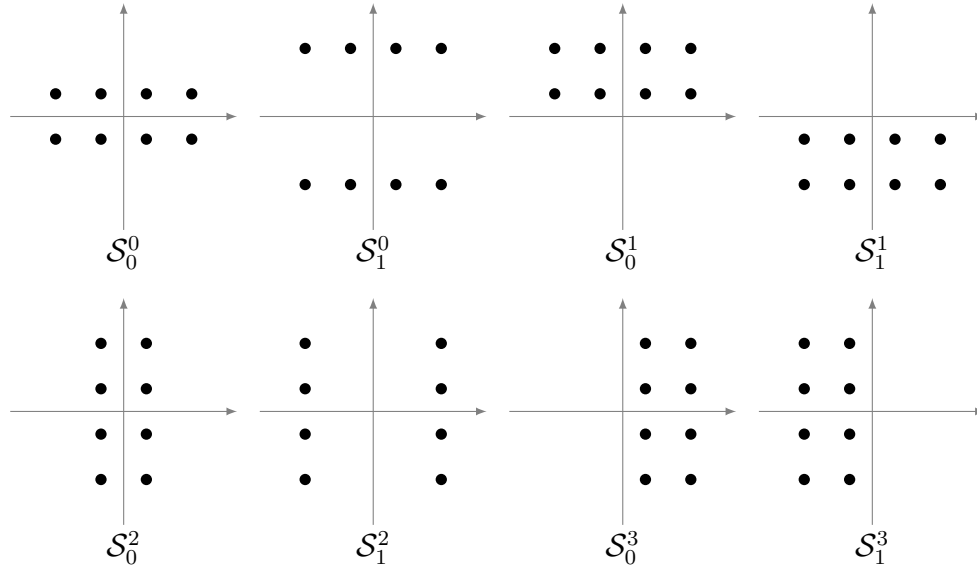


Figure 2.2: Above, 16-QAM constellation with Gray mapping. Below, subsets $\mathcal{S}_0^\ell$ and $\mathcal{S}_1^\ell$ for $\ell = 0, 1, 2, 3$ with the bits ordered from right to left.

With constellations such as BPSK and QPSK, where a single coded bit is mapped to the in-phase and quadrature dimensions of the constellation and thus **no dependencies among same-symbol coded bits**, we have that $\sum_k I(b_k; y) = I(\mathbf{s}; \mathbf{y})$, since the binary decoder is not disregarding information. Instead, **if the decoder disregards information** (acquiring information about other bits in the same symbol), the binary coding yields $\sum_k I(b_k; y) < I(\mathbf{s}; \mathbf{y})$.

Skipped

Several examples have been skipped: 1.20, 1.21, 1.22, 1.23

Anyhow, it's true that the difference between $\sum_k I(b_k; y)$ and $I(\mathbf{s}; \mathbf{y})$ is generally tiny, if the mapping of coded bits to constellation points is chosen wisely. An example of wise choice is **Gray mapping**, where neighboring constellation points only differ of one bit.

There's just one missing thing in order to have a full information-theory representation of modern transmission systems, which is **interleaving**, which generally concerns shuffling either symbols or bits in an invertible manner. Although symbol-level interleaving would suffice to break off bursts of poor channel conditions, bit-level interleaving has the added advantage of shuffling also the bits contained in a given symbol; if the interleaving were deep enough to push any bit dependencies beyond the confines of each codeword, then the gap between $\sum_k \mathcal{I}(b_k; y|\mathbf{s})$ would be closed. It's true that interleaving has no effect when $N \rightarrow \infty$ and will thus have no effect on the mutual information calculations, but it's also true that it will improve the performance of actual codes with finite N .

The usage of **binary coding/decoding, bit-level interleaving, constellation mappers/soft demappers** gives the so-called **Bit-Interleaved Coded Modulation (BICM)** architecture. BICM is information-theoretically equivalent to signal-space coding for

BPSK and Gray-Mapped QPSK. For higher-order constellations, even if the dependencies among same symbol bits are not fully eradicated by interleaving and the receiver ignores them, it is only slightly inferior.

In class:

What we usually do is that we have the input bits and we do **channel coding**; normally, channel coding is a **binary coding**.

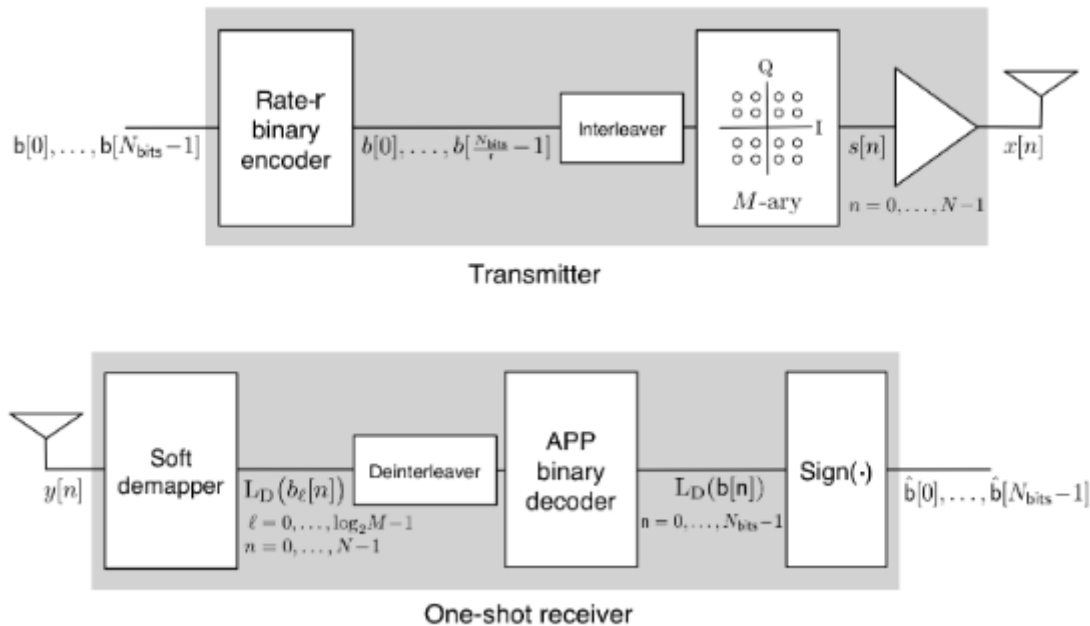


Figure 2.3: Top: BICM Transmitter Chain. Bottom: BICM Receiver Chain with Soft Demapper and Decoder.

Interleaver: sometimes the channel does errors in bursts, (it might be in a bad condition, ...); in that situation, the decoder cannot work as we'd have a long string of errors. What we do is to scramble the bits; if we have a matrix of memory, we write the values in rows

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

We read:

$$\begin{aligned} &\longrightarrow 1, 2, 3, 4, \\ &\longleftarrow 8, 7, 6, 5, \\ &\longrightarrow 9, 10, 11, 12, \\ &\longleftarrow 16, 15, 14, 13 \end{aligned}$$

And then:

↓ 1, 5, 9, 13,
↓ 2, 6, 10, 14,
↓ 3, 7, 11, 15,
↓ 4, 8, 12, 16

This way, bits which were close are now far apart, avoiding many consecutive errors; this is a simple interleaver! Then of course we'll do the reverse in the output.

There are many ways to implement interleavers, of course.

A positive **L-value** indicates that the bit in question is more likely to be a 1 than a 0, and vice versa for a negative value, with the magnitude indicating the confidence of the decision.

For a certain bit b , we define the **L-value** as

$$L(b) = \log \frac{P(b=1)}{P(b=0)}$$

such that if $L(b) > 0$, it's more likely that we have $b = 1$ and if $L(b) < 0$ it's more likely that we have $b = 0$. The bigger the magnitude, the more certain we are about our result.

At page 25 of the book, this concept is applied to an actual message, in the form of a **soft decoder**.

This thing we've just described (together with many other much more complex things handled from page 25 to page 30) is what is called an "A Posteriori Probability" decoder or "**APP**" decoder for short.

page 28

Whichever the type of code, the sign of the L-values generated by an APP decoder for the message bits directly gives the MAP decisions

APP decoders are indeed really cool.

Let's say we transmit bits; we have a system which gives us values between -1 and 1; the closer we are to 1, the more we're sure that the value will be 1; the closer we are to -1, the more we're sure the value will be 0. We then use the sign function and we have back the binary values.

From what I gathered, an **interleaver** is such to scramble bits in order to remove dependency between bits.

The rest: didn't get it at all :D

2.7 Spectral efficiency for finite-length codewords

We can reach the Shannon bound only by pushing N to infinity; only if we're sending a *really* long codeword can we reach the capacity. We'd need to wait a long time (an *infinite* amount of time, I think) to fill that codeword and then we'd have to wait lots of latency in order to receive such a large signal.

We can show that by using small (non infinite!) N values, we don't reach the capacity and we can indeed show to which spectral efficiency we actually get!

$$\frac{R}{B} = C - \sqrt{\frac{V}{N}} Q^{-1}(p_e) + \mathcal{O}\left(\frac{\log N}{N}\right). \quad (1.143)$$

How below we stay depends on N, V, p_e ; we don't need to remember the capacity, but we do need to remember the fact that **it's lower when compared to the capacity**.

Takeaway points are:

1. If the **block length N increases**, we get closer to the **ideal** capacity
2. If the **probability of error decreases**, we get closer to the **ideal** capacity (since $p_e \downarrow \implies Q \uparrow \implies (1/Q) \downarrow$)
3. If the **information density has a lower variance** (totally intuitively, i.e. if input/output are more related), we get closer to the **ideal** capacity

V is the variance of the information density, defined as

$$V = \text{var}[i(s; y)]. \quad (1.144)$$